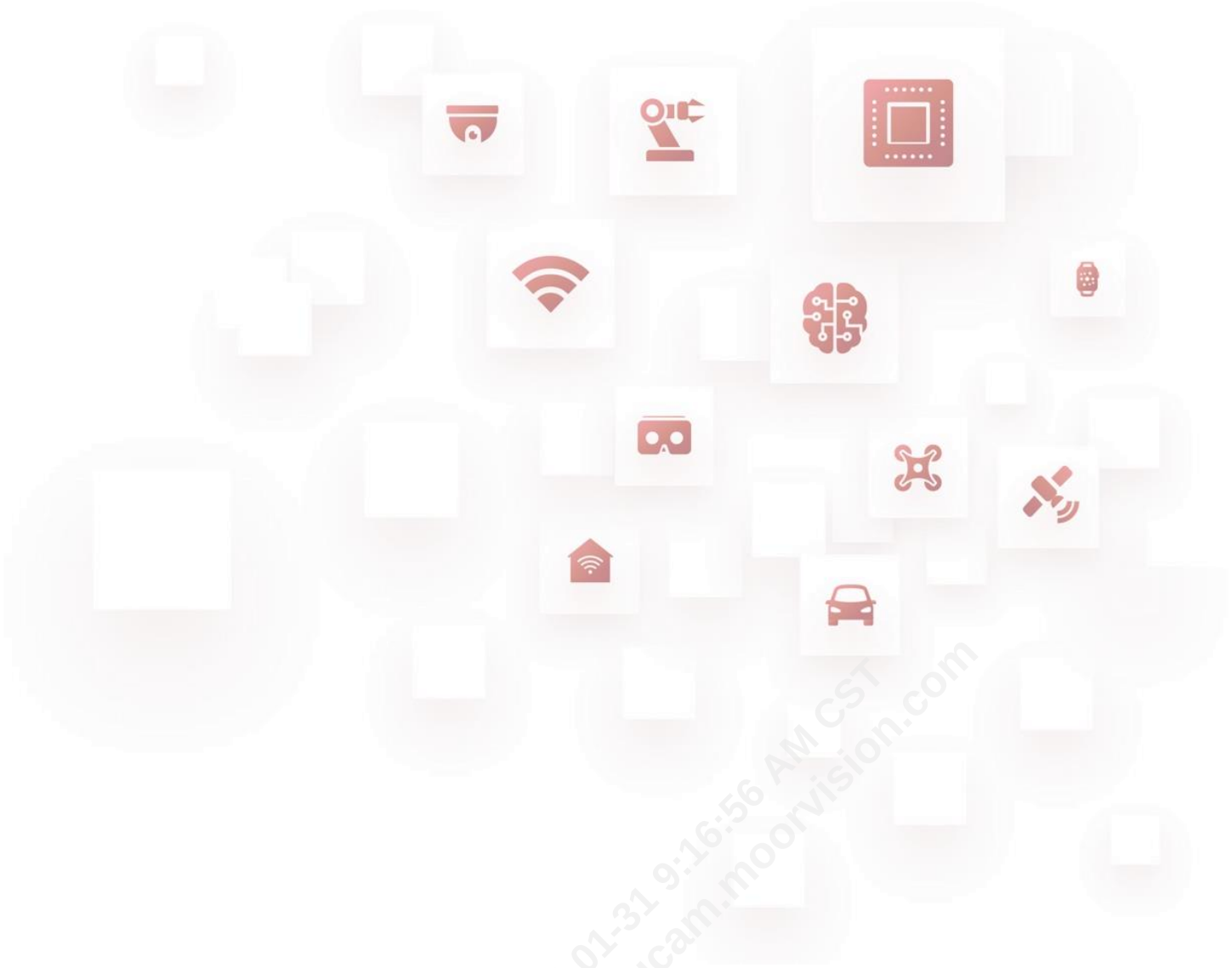
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Thundercomm TurboX C410/C610 Open Kit Linux Embedded

Software Release Notes

Rev V1.0
July 25, 2022

DN: tc-gle-2410
SDK Version: LE1.0.CS.r002007.1

Revision History

Revision	Date	Description
1.0	July 25, 2022	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on detailed steps to obtain, build, and program QCS610 LE (Linux Embedded) SDK (Software Development Kit) into the TurboX C410/C610 open kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The LE SDK supports the Linux operating system on the Thundercomm TurboX SOM. The LE SDK aims to provide a simple and flexible development platform to rapidly build IoT (Internet of Things) products with the QTI chipsets. The LE SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LE SDK is based on the Yocto Project 2.6 release and follows several of its development principles. The Yocto Project is used by the LE SDK to build framework.

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Chapter 2. System Requirements

Table 2-1 lists the hardware and software required to install and run the LE SDK.

Table 2-1. Hardware and software requirement to install and run the LE SDK

Hardware / Software Requirements	Source / Vendor	Function
Standalone hardware system minimum requirement: • 8 GB RAM • Quad core CPU ➲ **NOTE:** System with lower hardware specifications will increase compilation time. About an hour is the compilation time for a system with 16 GB RAM and Intel i7-2600 @3.4 GHz.	-	SDK build machine
Ubuntu 16.04 Long Term Support (LTS) Linux distribution for 64 bit architecture	Ubuntu Community/ Canonical, Ltd.	SDK build host Operating System (OS)
Repository (Repo)	SDK Open Source Project	SDK source management tool
Python 2.7.12 with sqlite3 module	Python.org	Building subsystem

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Chapter 3. Development Environment Setup

This chapter introduces the setup of a Linux development environment on an Ubuntu host machine.

Prerequisites:

Ensure your building host meet the following requirements:

- Host machine Linux distro – 16.04 LTS or 18.04 LTS
- Minimum 300 GB free disk space
- Tar v1.27 or newer
- Privileges to install utilities
- Install the host packages

3.1. Ubuntu 16.04 /18.04(64-bit) installation

Step 1. Install Ubuntu 16.04 /18.04to the computer following the instructions at:
http://www.ubuntu.com/download/desktop/install-ubuntu-desktop.

Step 2. To update the software, execute the following command:

```
$ sudo apt-get update
```

3.2. Tools and USB driver installation

This section provides instructions on installing **ADB** (Android Debug Bridge) (V1.0.41 or later), **Fastboot**, and USB drivers to your host computer.

The **Fastboot** is a tool used to flash images to the board by communicating with the boot loader.

The **ADB** is a tool installed on the host computer to access the device Unix shell, then help to push, modify, debug, and tweak device system files from the host computer.

3.2.1. ADB and Fastboot tools installation

To install **ADB** and **Fastboot**, execute the following commands.

```
$ sudo apt-get install android-tools-adb
$ sudo apt-get install android-tools-fastboot
```

3.2.2. USB driver installation and configuration

To install and configure USB drivers, follow the steps below:

Step 1. Set up the udev rules on a Linux computer.

1) Login as **root/sudo**.

```
$ cd /
```

2) Navigate to the directory */etc/udev/rules.d/*.

```
$ cd /etc/udev/rules.d/
```

3) Create the file *99-android.rules.*

```
$ sudo touch 99-android.rules
```

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4) Enter the following command get edit permission to file *99-android.rules*.

```
$ sudo chmod 777 /etc/udev/rules.d/99-android.rules
```

5) Enter *99-android.rules* by running the following command.

```
$ vim 99-android.rules
```

6) Input `i` to enter edit mode and add the following content to the file.

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", MODE="0777", GROUP="adm"
```

7) Press **ESC** key to exit the edit mode.

8) Enter `:wq` to save the change and exit.

Step 2. Restart the udev service by executing the following commands.

```
$ sudo chmod a+r /etc/udev/rules.d/99-android.rules
$ sudo service udev restart
```

Step 3. Connect the board to your computer with an USB cable, and power on the target board to start the system. Wait for several seconds then input `$ adb devices`. The device is correctly detected with the following information displayed on your screen.

```
$ adb devices
List of devices attached
7050372c       device
```

➲ **NOTE:** Avoid connecting the device to the PC with a USB hub. Connect the device to the PC directly via a USB cable.

3.3. Required Building Environment

Follow the steps below to configure the building environment by install the proper version GCC/G++ and Linux package dependencies.

3.3.1. Java 8

Step 1. Check the version. Run the following commands to ensure Java 8 is installed.

```
$ java -version
openjdk version "1.8.0_131"
OpenJDK Runtime Environment (build 1.8.0_131-8u131-b11-0ubuntu1.16.04.2-b11)
OpenJDK 64-Bit Server VM (build 25.131-b11, mixed mode)
```

Step 2. Update the version.

With version 1.7 preinstalled, run the following commands to upgrade it to version 1.8:

```
$ sudo apt-get update
$ sudo apt-get install openjdk-8-jdk
```

➲ **NOTE:** The Java 8 should be used along with Ubuntu 16.04.

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3.3.2. Linux package dependencies

Step 1. Install the dependency libraries to the host computer.

```
$ sudo apt-get install git-core gnupg flex bison build-essential zip curl zlib1g-dev
gcc-multilib g++-multilib libc6-dev-i386 lib32ncurses5-dev x11proto-core-dev libx11-
dev lib32z1-dev libgl1-mesa-dev libxml2-utils xsltproc unzip fontconfig libxml-
simple-perl texinfo gawk chrpath
```

Step 2. Set **bash** as default shell tool.

```
$ sudo rm /bin/sh
$ sudo ln -sf /bin/bash /bin/sh
```

➲ **NOTE**: For more information on building environment dependencies, refer to AOSP documents at: *https://source.android.com/setup/build/initializing*.

3.3.3. Install repo

Run the following command to install repo.

```
$ sudo apt-get install repo
```

3.3.4. Configure git global variable

Configure user name and email.

```
$ git config --global user.email "<User Email>"
$ git config --global user.name "<User Name>"
```

➲ **NOTE:** Replace the `<User Email>` and `<User Name>` with the email address and user name you have registered at *Thundercomm Website*.

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Chapter 4. Get Started with LE SDK

4.1. LE SDK information

Table 4-1. LE SDK Information

SDK Version	Baseline	Baseline Distribution	OS Version
LE1.0.CS.r002007.1	QCS610.LE.1.0	CS3.8 0.0.130.2	Linux Embedded 1.0

4.2. Block diagram

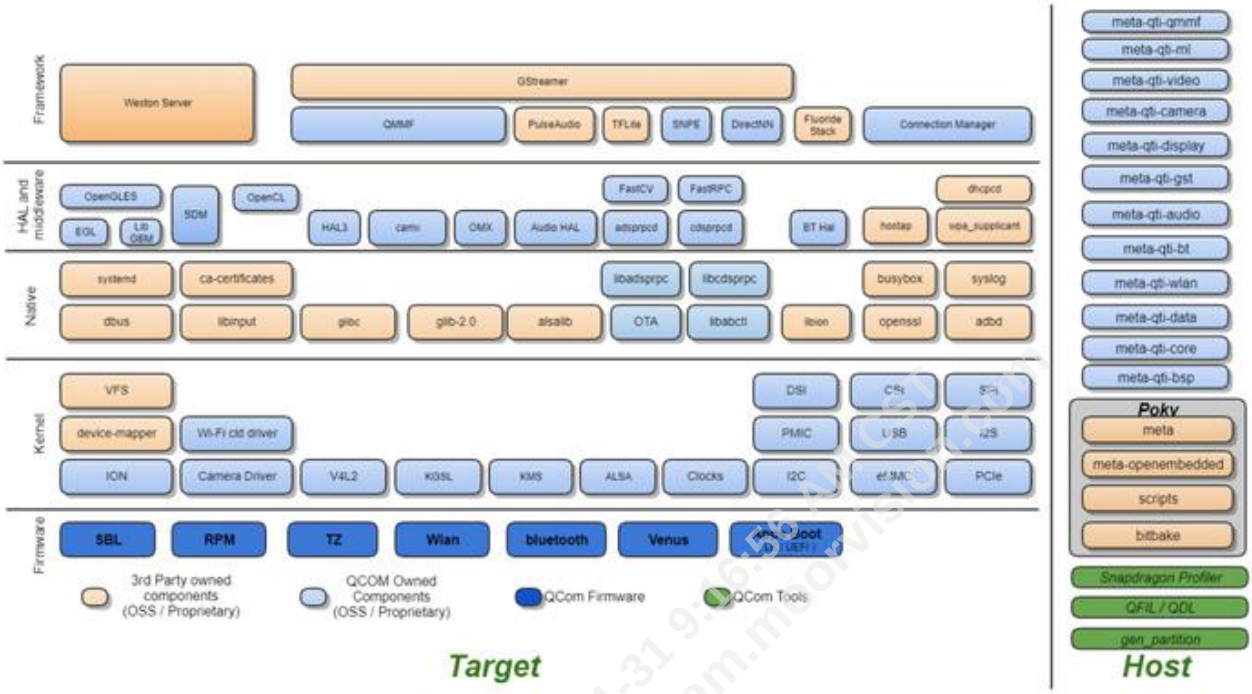


Figure 4-1.Block Diagram

4.3. SDK source tree structure

```
SDK.Turbox_C610.LE1.0_CS/
├── about.html -> common/about.html           //SDK version info(Product,
Distribution,Components, …)
├── adsp_proc             //aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              //AOP image is released as a binary and no build compilation is
needed
├── apps_proc
   ├── build-qti-distro-fullstack-virtualization-debug           /* Directory of temporary
                                                    fileand image files generated by
compilation */
   ├── disregard
   ├── downloads                          //Software packages required for compilation
   ├── poky                          //Yocto's recipe directory
   ├── prebuilt_HY11                     //Qcom privatized software package
   ├── setup-environment -> poky/qti-conf/set_bb_env.sh    //Script for set the build
environement
   ├── src                           /* AP source code, such as linux kernel, filesystem,
vendor,                                                                  OTA, etc. */
   ├── sstate-cache
   └── syncbuild.sh
```

```
├── boot_images           //BOOT image is released as a binary and no build compilation is
needed
├── BP-CODE                    //BP binary files
├── btfm_proc             //BTFM image is released as a binary and no build compilation is
needed
├── cdsp_proc             //cDSP image is released as a binary and no build compilation is
needed
├── common/config/emmc/partition.xml                          eMMC partition file
├── contents.xml -> common/contents.xml              //image contents file
├── modem_proc            //MODEM image is released as a binary and no build compilation
is needed
├── trustzone_images      //TZ image is released as a binary and no build compilation is
needed
├── turbox
│   └── tools
│       ├── build_script                        //SDK build script(android build, zip, …)
│       └── flash_tools                              //Flash tools files(qdl, …)
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh       //SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh       //Fastboot
flash tool
├── venus_proc            //VIDEO image is released as a binary and no build compilation
is needed
├── wdsp_proc             //wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             //WLAN image is released as a binary and no build compilation is
needed.
```

4.4. Download LE SDK

Follow the steps below to activate an account and download the SDK code.

4.4.1. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
$ echo "machine partner.thundercomm.com login <User Email> password <password>" >> ~/.netrc
```

➲ **NOTE:** Please use the email and password you have registered at www.thundercomm.com to replace `<User Email>` and `<password>` in the above command.

4.4.2. Download the LE SDK code

Follow the steps below to download the LE SDK:

Step 1.Initialize the repository.

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.turbox-c610-
le1.0-dev -m turbox-c610-le1.0-v2-dev.release.CS.r002007.1.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

Step 2.Synchronize the source code.

```
$ repo sync -c -j8 --no-tags
```

4.5. SDK version update

If you have downloaded the old-version SDK, there are two ways to download the new SDK fast.

Option 1. Upgrade the local old folder to a new release

Navigate to the old local code folder. You can run the repo init command of new release as *4.4.2.Download the LE SDK code*, then run `repo sync -j4 -c --no-tags` to receive the new code.

Option 2. Download the new SDK source code to a new folder (recommended)

Create a new folder, and run `repo init` with `-reference` value.

For example:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.turbox-c610-le1.0-dev -m turbox-c610-le1.0-v2-dev.release.CS.r002007.1.xml --repo-url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-repo-verify --reference=<old_sdk_code_path>
$ repo sync -j4 -c --no-tags
```

➲ **NOTE:** In `--reference=<old_sdk_code_path>`, the `old_sdk_code_path` is the absolute path of your old SDK folder. For example:

```
--reference=/home/host/turbox-c610-le1.0-v2-dev.release.CS.r002006
```

4.6. Build image

To compile the AP code, run the following script file according to your image version in the root directory.

- Build debug version:

```
$ ./turbox_build.sh -a -l -v debug
```

- Build perf version:

```
$ ./turbox_build.sh -a -l -v perf
```

➲ **NOTE**: UART debug cannot be used in perf/user version.

- Run the following commands to rebuild *kernel* additionally (if necessary).

```
$ export SHELL=/bin/bash
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-virtualization-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc make_bootimg qti-multimedia-image
```

➲ **NOTE**: The default *boot.img* has verify enabled and cannot be flashed unless the kernel is rebuilt. Hence, when rebuilding only the kernel, you need to flash the *boot.img.noverity* which has verify disabled, refer to *turbox_flash.sh* to flash the *boot.img.noverity*.

- Run the following commands to rebuild *dtbo* additionally (if necessary).

```
$ export SHELL=/bin/bash
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-virtualization-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc cleanall linux-msm && bitbake linux-msm
```

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4.7. Pack ROM package

Step 1. Run the following command to update common information according to your image version.

- **Debug version:**

```
$ ./turbox_build.sh -u --uci-imf -v debug
```

- **Perf version:**

```
$ ./turbox_build.sh -u --uci-imf -v perf
```

➲ **NOTE:** The command is successfully executed with the following log displayed.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the packed ROM package at *turbox/output*. Replace `<debug/perf>` with debug or perf.

- Pack Fastboot image

```
$ ./turbox_build.sh --zip_fastboot_build -l -v <debug/perf>
```

- Pack FlatBuild image

```
$ ./turbox_build.sh --zip_flat_build -l -v <debug/perf>
```

➲ **NOTE:** Customers can also get the ROM package provided by Thundercomm from:

- **For debug version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE1.0.R.debug.CS.r002007.1.zip*

    md5: ca251fd06992eca145f2be9dfd972c9c
  - *FastbootBuild_TurboX_C610_xx.xx_LE1.0.R.debug.CS.r002007.1.zip*

    md5: 6920b675e6a75639e5a3484e4abcdf4c
- **For perf version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE1.0.R.perf.CS.r002007.1.zip*

    md5: b0cd961f6d040120ae41c9c5d95fb3a1
  - *FastbootBuild_TurboX_C610_xx.xx_LE1.0.R.perf.CS.r002007.1.zip*

    md5: 95157a497ba86448d50f63b2d1e76f1c

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4.8. Flash images to board

➲ **NOTE**: If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with *turbox_flash_flat.sh*.

4.8.1. Flash Fastboot image

Step 1. Enter the **Fastboot** mode. Two options are available for your selection:

Option 1. Run the following commands when the device is in ADB mode.

```
$ adb root
$ adb reboot bootloader
```

Option 2. Alternatively, you can proceed with the steps below to enter the **Fastboot** mode.

1) Power off the device.

2) Hold the **Volume down** button and press the **Power on** button for about 5 seconds.

3) Release the **Volume Down** button.

To check if the device has entered the **Fastboot** mode, run the following command:

```
$ fastboot devices
```

The process is successful with the following output displaying on the shell:

```
7050372c fastboot
```

Step 2. Flash images.

Execute the commands below to flash images to the board. Replace **<folder name of image package>** with the actual image package name of your product.

```
$ cd <folder name of image package>
$ bash ./turbox_flash.sh -r -v <debug/perf>
```

➲ **NOTE**: Enter **./turbox_flash -h** to acquire detailed usage.

```
$ ./turbox_flash -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  The images except gpt,userdata,persist will be flashed if no options input
Description:
  flash images with fasboot
Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
  -f, --force gpt,persist and userdata will be flashed.
              User data and calibration data(BT/WIFI MAC address, etc..) will be
erased.
  -r, --reboot      reboot the device automatically after flashing completed.
  -b, --nonhlos     flash all bp images(xbl,rpm,ftfm,tz,dsp,modem,pmic).
  -a, --hlos       flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp).
  --all       flash all images
Example:
  ./turbox_flash.sh
     Flash the images except gpt,userdata,persist.
  ./turbox_flash.sh --force
     Flash all images.
  ./turbox_flash.sh --hlos
     Flash all ap images.
```

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4.8.2. Flash FlatBuild image

Follow the steps below to flash FlatBuild image.

Step 1. Download the flashing tool: *turbox_flash_flat.sh*.

Step 2. Flash the device

1) Prepare the FlatBuild images.

Check if the images are ready, then copy the *turbox_flash_flat.sh* to image folder.

Replace **<folder name of image package>** with the actual image package name of your product.

```
$ cp turbox_flash_flat.sh <folder name of image package>
$ cd <folder name of image package>
$ chmod +x turbox_flash_flat.sh
```

2) Enter **9008 Mode**.

a. Power off the the device by disconnecting the power cable and USB cable.

b. Connect the power cable.

c. Hold the **Force USB Boot** button.

d. Press the **Power on** button and wait for about 3 seconds.

e. Release the **Force USB Boot** button and **Power on** button.

f. Connect the Type-C USB cable to an Ubuntu computer.

To check if your device has entered the **9008 Mode**, run the following command:

```
$ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
…… .
```

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3) Start to flash the FlatBuild images.

Run the following command under the image folder.

```
sudo ./turbox_flash_flat.sh -x patch0.xml,rawprogram_unsparse0.xml -
f ./prog_firehose_ddr.elf -i ./ -s eMMC
```

If the flashing process is successful, the following log will be printed.

```
15:41:09: INFO:      _            (done)
15:41:09: INFO:     | |
15:41:09: INFO:   __| | ___  _ __   ___
15:41:09: INFO:  / _` |/ _ \| '_ \ / _ \
15:41:09: INFO: | (_| | (_) | | | |  __/
15:41:09: INFO:  \__,_|\___/|_| |_|\___|
15:41:09: INFO: {All Finished Successfully}

15:41:09: INFO: Overall to target 201.617 seconds (20.88 MBps)
15:41:09: INFO: {percent files transferred 100.00%}
```

4) Reboot the device.

a. Power off the device by disconnecting the power cable and USB cable.

b. Connect the power cable to the board.

c. Hold the **Power on** button and wait for about 3 seconds.

d. Connect the Type-C USB cable and check the adb device with the following command:

```
$ adb devices
List of devices attached
b237534       device
```

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Chapter 5. Features

5.1. Presented features

This section describes the key software features that have been included in earlier versions.

- **SDK**
  - Yocto Thud (v2.6)
  - GCC toolchain (v8.2.0)
  - glibc (v2.28)
  - systemd (v1.239)
  - OpenEmbedded Linux
- **Platform**
  - Linux Kernel 4.14
  - 64-bit only
  - Azure and Docker
  - UBWCf
  - OTA
  - HY22
  - DTB Overlay
- **CV**
  - FastCV
  - OpenCV (v3.4.3)
- **Security**
  - DM-Verity
  - SELinux
  - KASan
  - FDE
  - VBLE
  - Non-Root Processes
  - Linux Full Disk encryption
- **Connectivity**
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
  - Wi-Fi support WPA3 in AP mode
- **Storage**
  - eMMC
- **Multimedia**
  - GStreamer Framework
  - Hexagon-SDK

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- Dynamic image rotation (with GStreamer) via GPU
- Static image rotation (with GStreamer) via VPU before recording

- **Camera**
  - Staggered HDR (v2&v3.8)
  - IRLED
  - LDC and EIS
  - Dual Independent Cameras
  - Force sensor mode to bypass default logic
  - Manual disablement of IR cut filter with IR LED OFF
- **Video**
  - JPEG DMA Engine
  - H264 & HEVC encode and decode
- **Audio**
  - AAC, PCM and MP3 (only playback) Encode and Playback
  - Audio + Hands Free Profile concurrency
- **Display**
  - DP
  - LCD
- **Graphics**
  - OpenGL ES 3.2
  - OpenCL 2.1
- **AI**
  - BENCHMARK APP
  - AI Inferencing using SNPE v1.36
  - AI Inferencing using TFlite 2.0
  - Tflite supports NNAPI delegate offloading graph processing to DSP
  - SNPE run on GPU CPU and DSP
- **Battery**
  - Software-based fuel gauge solution for battery powered devices

5.2. New features

None.

5.3. On-going features

None.

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Chapter 6. Test Report

Table 6-1. Test report

Test Case	Result
Flatbuild version flash	Pass
Fastboot version flash	Pass
Metabuild version flash	Pass
Boot_adb reboot	Pass
Boot_adb reboot bootloader	Pass
Boot_fastboot reboot	Pass
Fastboot (adb & volume down)	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
BT/Wi-Fi/59/Red/Green/Blue LED	Pass
LSM6DSOTR sensor	Pass
Memory Information	Pass
Power suspend	Pass
Incremental-OTA	Pass
64-bit user space	Pass
Disable verity	Pass
SD Card auto mount/unmount/multiple plug & unplug	Pass
SDcard file push/pull	Pass
Multimedia_GST_recorder_4k_30_AVC_MP4_mux	Pass
Multimedia_GST_recorder_4k_30_HEVC_MP4_mux	Pass
Multimedia_GST_recorder_1080p_30_AVC_filesave	Pass
Multimedia_GST_Snapshot_One 1080p AVC mp4 muxed video with JPEG snapshots	Pass
Multimedia _GST_Overlay_1080p_30 YUV with UserText overlay	Pass
Multimedia _GST_Display_1080p_30 Live preview on a local display	Pass
Multimedia _GST_Display_1080p_30 Live preview over TCP/RTSP	Pass
Camera_LDC/ ZOOM	Pass
Camera_Dual Independent Cameras (IMX577)	Pass
Audio line in	Pass
Audio_1-6 D-MIC recorder individually	Pass
Audio_1-6 channels recorder concurrency with GST	Pass
Audio_1-6 channels recorder concurrency with parecord	Pass
Audio headphone	Pass
Audio speaker	Pass
Display_DP	Pass

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Test Case	Result
Display_LCD	Pass
Display_Weston Color Bar	Pass
USB_Type-C functionality	Pass
USB_Type-A functionality	Pass
USB_Micro for UART	Pass
Ethernet_RJ45 connect	Pass
Ethernet_ Ping test	Pass
Connectivity_Bluetooth	Pass
Connectivity_ Wi-Fi_connect to AP	Pass
Battery	Pass
Charge the battery via type-c	Pass
12V DC power supply	Pass
HDMI IN	Pass
AI inferencing via TFlite 2.0	Pass

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Chapter 7. Known Issues

The following issues have been fixed in this release:

- Optimize compatibility of HDMI in input source.
- Optimize RF parameters.

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